Steering Vectors Differ Across Personas

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Key Finding: Steering vectors for the same trait encode qualitatively distinct representations depending on the active persona. They differ in both magnitude and direction (cosine similarity). RLHF-trained traits like honesty show reduced persona diversity, suggesting training collapses representational variation.

Motivation

Character traits like sycophancy correspond to linear directions in activation space — persona vectors — that can monitor and steer model behaviour (Chen et al., 2025).

The Assistant Axis captures how "assistant-like" a model is (Lu et al., 2026). Both extract steering vectors from a single baseline persona.

But models are deployed across diverse operational modes, each a distinct persona state. Do steering vectors transfer across personas, or do personas fundamentally reshape trait representations?

Method

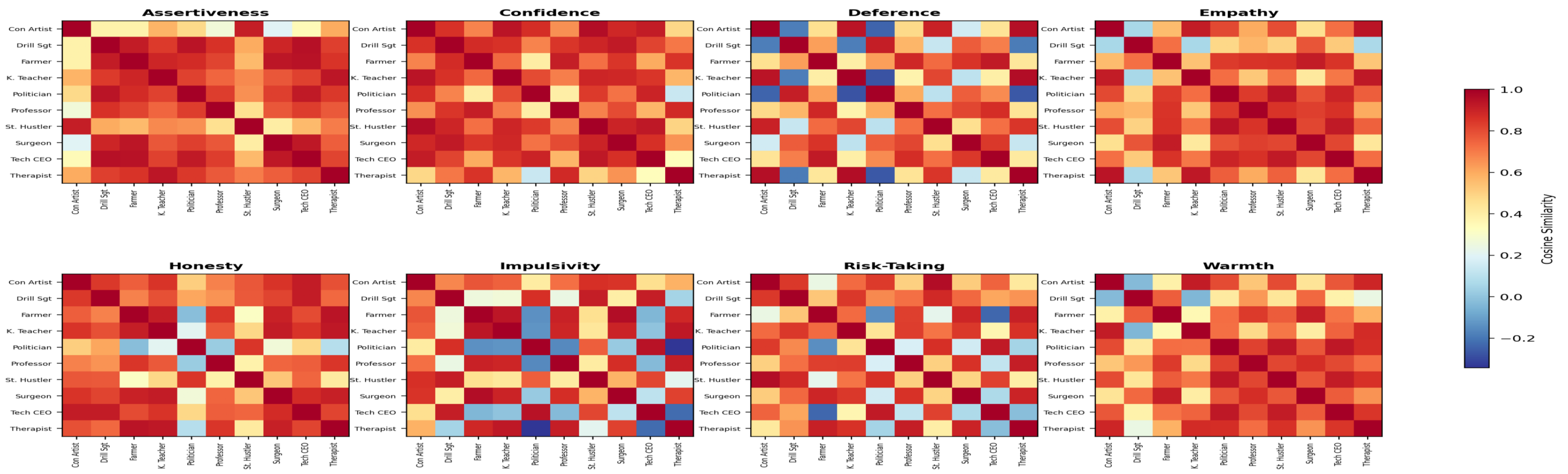
10 persona archetypes induced via system prompts (Farmer, Politician, Therapist, Drill Sergeant, Street Hustler, Professor, Tech CEO, Kindergarten Teacher, Surgeon, Con Artist).

8 traits: assertiveness, empathy, risk-taking, honesty, confidence, deference, warmth, impulsivity.

Contrastive activation addition: 5 pos/neg instruction pairs × 20 questions = 100 contrastive pairs per persona–trait. Same question under opposing instructions isolates the trait signal.

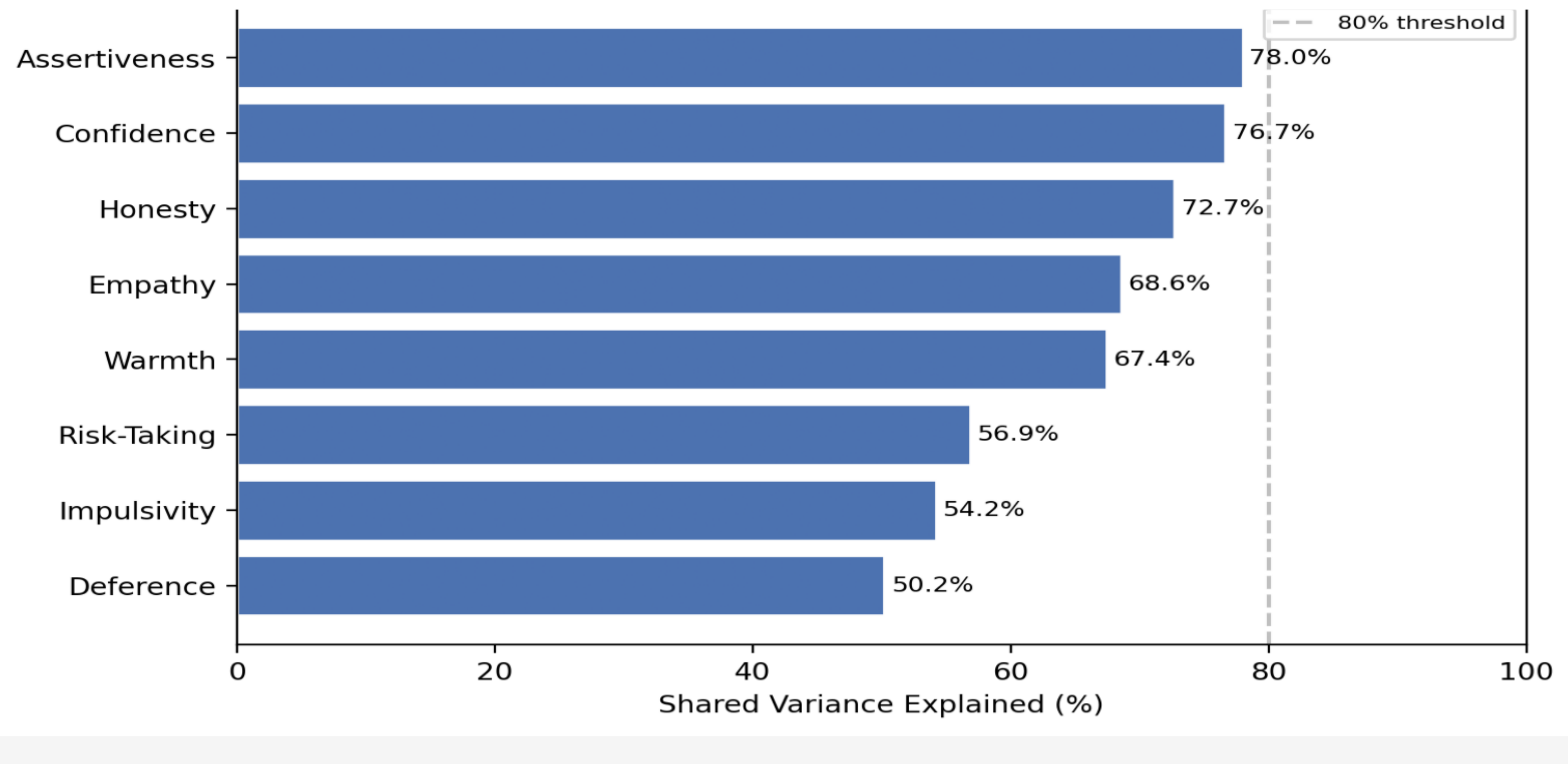
Geometric Similarity: Cross-Persona Cosine Similarity of Steering Vectors

Each heatmap shows pairwise cosine similarity between persona-specific steering vectors for one trait. Red = similar encoding, Blue = distinct representations.



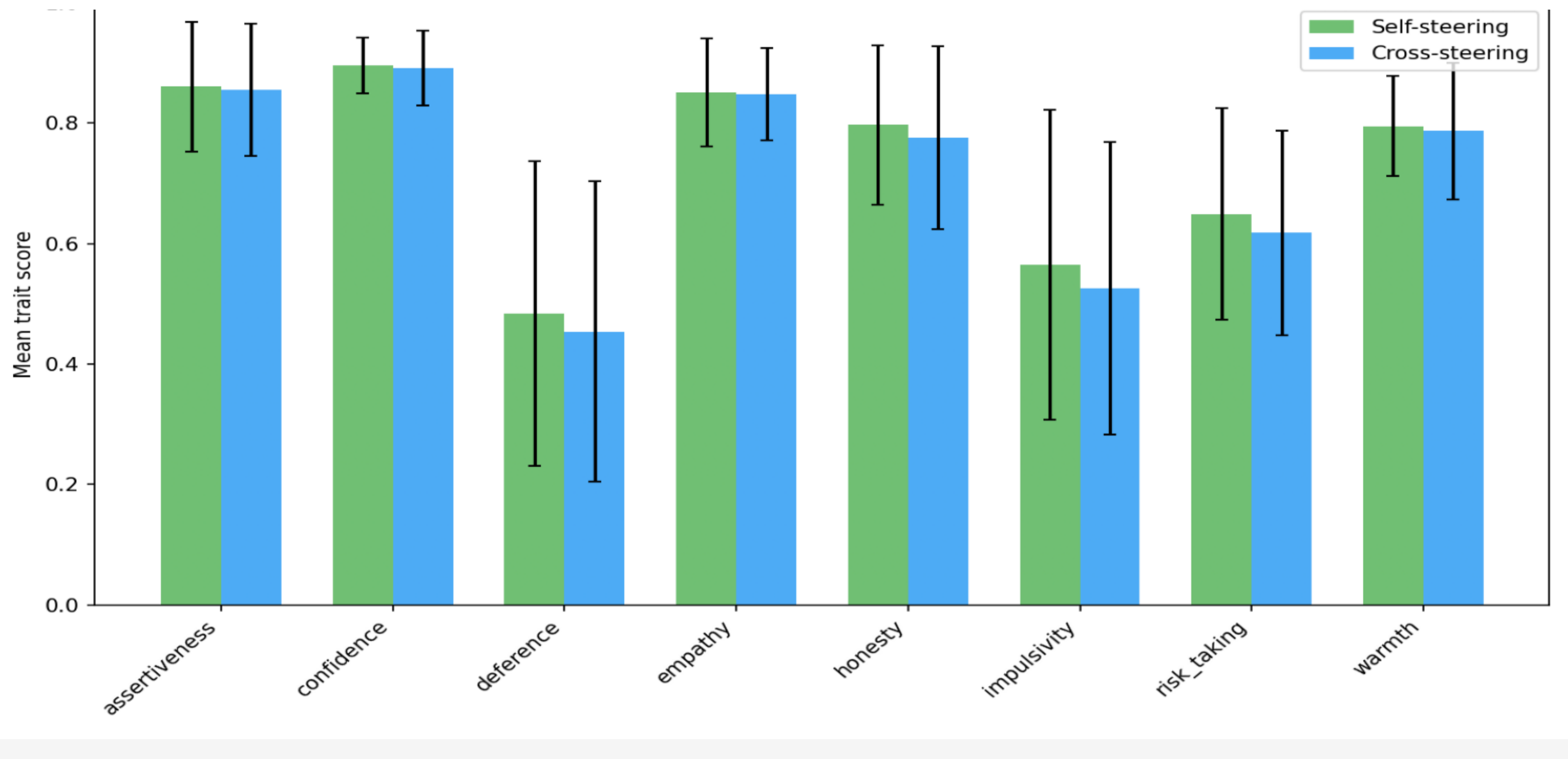
Shared vs Persona-Specific Variance

No trait exceeds 80% shared variance — all encode substantial persona-specific signal.



Behavioural Validation: Self vs Cross Steering

Self-steering slightly outperforms cross-persona, confirming vectors are persona-conditioned.



Key Observations

- Steering vectors differ drastically across personas — the same trait is encoded differently depending on identity.
- Traits heavily optimized during RLHF (e.g. honesty) show higher cross-persona similarity, suggesting training collapses representational diversity.
- Intuitively similar persona–trait pairs share similar vectors (e.g. kindergarten teacher ≈ therapist on empathy), while unexpected pairs diverge (tech CEO ≠ surgeon on risk).
- Cross-persona steering introduces "residues" — biases and quirks from the source persona leak into the target.

Implications for Safety

Safety and alignment work implicitly assumes steering vectors are model-global — that a sycophancy suppression vector extracted from the default assistant works equally for a chat assistant, an autonomous agent, or a roleplayed character.

Our results challenge this assumption: persona-specific representations mean that a single intervention may not generalise across deployment contexts.

If trait representations are persona-conditioned, safety teams may need persona-aware interventions rather than one-size-fits-all steering vectors.

Next Steps

- Track steering effect across the training pipeline (base → SFT → DPO → Instruct)
- Extend to safety-critical traits: sycophancy, refusal, power-seeking
- Measure trait coupling — side-effects of single-trait steering on other traits
- Validate on additional model families (OLMo, Llama)

References

Chen, R., Arditi, A., Sleight, H., Evans, O., & Lindsey, J. (2025). Persona Vectors: Monitoring and Controlling Character Traits in Language Models. arXiv:2507.21509.

Lu, C., Gallagher, J., Michala, J., Fish, K., & Lindsey, J. (2026). The Assistant Axis: Situating and Stabilizing the Default Persona of Language Models. arXiv:2601.10387.